6	**(a)**	$^{234}_{94}Pu \rightarrow {}^{234}_{92}U + {}^{4}_{2}He$	A1
	(b)(i)	Applying conservation of mass-energy, $(m_{Pu} – m_U – m_\alpha)\ c^2$ = energy released = E_k of the products $(238.0496 – m_U – 4.0026) \times (1.66 \times 10^{-27})\ c^2 = 5.649\times10^6 \times (1.60\times10^{-19})$ $m_U = 234.0410$ u (4 dp)	 C1 C1 A1
	(b)(ii)	By the principle of conservation of momentum $m_{Pu}v_{Pu} = m_u v_u + m_\alpha(-v_\alpha)$ $0 = m_U v_U - m_\alpha v_\alpha$ $\frac{v_\alpha}{v_U} = \frac{m_U}{m_\alpha}$ Ratio of $E_k = \frac{E_{K,\alpha}}{E_{K,U}} = \frac{\frac{1}{2}m_\alpha {v_\alpha}^2}{\frac{1}{2}m_U {v_U}^2}$ $= \frac{m_\alpha}{m_U}(\frac{m_U}{m_\alpha})^2$ $= \frac{m_U}{m_\alpha}$ $= \frac{234.0410}{4.0026}$ = 58.5	 M1 M1 A1 A0
	(b)(iii)	$E_{K,U} + E_{K,\alpha} = 5.649$ MeV $\frac{1}{58.5} E_{K,\alpha} + E_{K,\alpha} = 5.649$ MeV $E_{K,\alpha} = 5.55$ MeV	 C1 A1
	(c)(i)	Radioactivity is a spontaneous process that is not affected by any physical condition or the chemical combination in which the nucleus exists. Hence, the properties of the nucleus remain unchanged no matter what chemical compound it is found in.	B1
	(c)(ii)	Alpha particles are readily absorbed by a sheet of paper and cannot penetrate the outer, protective layers of skin. Hence, plutonium is not a hazard as long as it remains outside the skin. When the dust is inhaled into the lungs, the emitted alpha particles deliver all their energy to sensitive internal tissues and may cause damage.	 B1

7	(c)	(i)	vertical height from ball launcher	